3.4 Algebraic Topology — Study Notes

These notes condense Hatcher Chapter 0, Chapters 1–2 and §§3.1–3.2. They are a map and a revision aid, not a replacement for the text. Spaces are topological spaces, maps are continuous, and $I = [0, 1]$.

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1. Homotopy and CW complexes

1.1 Homotopy

A **homotopy** between maps $f_0, f_1 : X \to Y$ is a map $F : X \times I \to Y$ with $F(\cdot, 0) = f_0$ and $F(\cdot, 1) = f_1$. Write $f_0 \simeq f_1$. If $F(a, t)$ is independent of t for $a \in A \subseteq X$, the homotopy is **rel** A.

A map $f : X \to Y$ is a **homotopy equivalence** if there is $g : Y \to X$ with $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$. X is **contractible** if it is homotopy equivalent to a point.

A **retraction** $r : X \to A$ is a map with $r|_A = \mathrm{id}_A$. A **deformation retraction** is a homotopy rel A from id_X to a retraction onto A. If A is a deformation retract of X, the inclusion $A \hookrightarrow X$ is a homotopy equivalence.

Examples. $\mathbb{R}^n \setminus \{0\}$ deformation retracts onto S^{n-1} via $F(x, t) = (1-t)x + t\,x/|x|$. The Möbius band deformation retracts onto its core circle. A torus minus a point deformation retracts onto $S^1 \vee S^1$ (push out radially from the missing point in the square model).

Key idea. Every invariant in this course (π_1, H_n, H^n) is a functor that is *homotopy invariant*: homotopic maps induce the same homomorphism, so homotopy equivalent spaces have isomorphic invariants.

1.2 CW complexes

A **CW complex** is built inductively: X^0 is a discrete set; X^n is obtained from X^{n-1} by attaching n-cells e^n_α via maps $\varphi_\alpha : S^{n-1} \to X^{n-1}$, i.e. $X^n = \left(X^{n-1} \sqcup_\alpha D^n_\alpha\right)/(x \sim \varphi_\alpha(x))$; $X = \bigcup_n X^n$ with the weak topology (a set is closed iff it meets each closed cell in a closed set).

Standard examples:

Space	Cells
S^n	one 0-cell, one n-cell (attached by the constant map)
$\mathbb{RP}^n$	one cell in each dimension $0, \dots, n$; e^k attached by the 2-to-1 quotient $S^{k-1} \to \mathbb{RP}^{k-1}$
$\mathbb{CP}^n$	one cell in each even dimension $0, 2, \dots, 2n$
closed orientable surface M_g	one 0-cell, $2g$ 1-cells a_i, b_i, one 2-cell attached along $[a_1, b_1] \cdots [a_g, b_g]$
closed non-orientable N_h	one 0-cell, h 1-cells, one 2-cell attached along $a_1^2 \cdots a_h^2$

Homotopy extension property (HEP). A CW pair (X, A) (with A a subcomplex) has the HEP: any homotopy of A extends to X given an extension at time 0. Consequence: if A is a contractible subcomplex, then $X \to X/A$ is a homotopy equivalence. For example, collapsing a maximal tree in a connected graph shows every connected graph is homotopy equivalent to a wedge of circles.

Operations. Wedge sum $X \vee Y$, suspension SX, cone CX, smash product. $SS^n \cong S^{n+1}$.

2. The fundamental group

2.1 Definition

For $x_0 \in X$, $\pi_1(X, x_0)$ is the set of homotopy classes rel endpoints of loops $\gamma : I \to X$ with $\gamma(0) = \gamma(1) = x_0$, under concatenation $[\gamma][\delta] = [\gamma \cdot \delta]$. It is a group (identity: constant loop; inverse: reversed loop). A path h from x_0 to x_1 gives an isomorphism $\beta_h : \pi_1(X, x_1) \to \pi_1(X, x_0)$, $[\gamma] \mapsto [h \cdot \gamma \cdot \bar{h}]$, so for path-connected X the group is well defined up to isomorphism. X is **simply connected** if path-connected with trivial π_1.

A map $f : (X, x_0) \to (Y, y_0)$ induces $f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$; $(gf)_* = g_* f_*$; $\mathrm{id}_* = \mathrm{id}$. A homotopy equivalence induces an isomorphism (with care about basepoints: if f_t moves the basepoint along h, then $f_{0*} = \beta_h f_{1*}$).

2.2 $\pi_1(S^1) \cong \mathbb{Z}$

Theorem. $\pi_1(S^1, 1) \cong \mathbb{Z}$, generated by the loop $\omega(s) = e^{2\pi i s}$.

Proof sketch. Let $p : \mathbb{R} \to S^1$, $p(s) = e^{2\pi i s}$. Two lifting facts:

1. **Path lifting.** For each path γ in S^1 starting at $p(\tilde{x}_0)$ there is a unique lift $\tilde{\gamma}$ in $\mathbb{R}$ starting at $\tilde{x}_0$.
2. **Homotopy lifting.** A homotopy of paths γ_t lifts uniquely to a homotopy $\tilde{\gamma}_t$ of lifts with fixed starting point.

Both are proved by a Lebesgue number argument: cover S^1 by evenly covered open arcs, subdivide I (or $I \times I$) so each small piece maps into one arc, and lift piece by piece using the local homeomorphisms.

Define $\Phi : \pi_1(S^1, 1) \to \mathbb{Z}$ by $\Phi[\gamma] = \tilde{\gamma}(1)$ where $\tilde{\gamma}$ is the lift starting at 0. Homotopy lifting makes Φ well defined (endpoint is fixed along a homotopy rel endpoints, since it varies continuously in the discrete set $\mathbb{Z}$). It is a homomorphism since the lift of $\gamma \cdot \delta$ is $\tilde{\gamma}$ followed by $\tilde{\delta} + \tilde{\gamma}(1)$. Surjective: ω^n lifts to $s \mapsto ns$. Injective: if $\tilde{\gamma}(1) = 0$, $\tilde{\gamma}$ is a loop in the contractible $\mathbb{R}$, so $\tilde{\gamma} \simeq$ const rel endpoints, and composing with p gives $\gamma \simeq$ const. □

Applications.

- **No retraction** $D^2 \to S^1$: if r were one, $r \circ i = \mathrm{id}_{S^1}$ gives $\mathbb{Z} \to 0 \to \mathbb{Z}$ composing to the identity, impossible.
- **Brouwer (dim 2).** If $h : D^2 \to D^2$ has no fixed point, the ray from $h(x)$ through x hits S^1 at $r(x)$, a retraction. Contradiction.
- **Fundamental theorem of algebra.** If $p(z) = z^n + \ldots$ has no root, the loops $f_r(s) = p(re^{2\pi i s})/|p(re^{2\pi i s})|$ are all homotopic (vary r), trivial at $r = 0$, and homotopic to ω^n for large r; so $n = 0$.
- **Borsuk–Ulam (dim 2).** Every map $S^2 \to \mathbb{R}^2$ sends some pair of antipodal points to the same point.
- **Products.** $\pi_1(X \times Y) \cong \pi_1 X \times \pi_1 Y$; so $\pi_1(T^n) \cong \mathbb{Z}^n$.
- **Spheres.** $\pi_1(S^n) = 0$ for $n \geq 2$ (a loop can be homotoped to miss a point, and $S^n \setminus \{pt\} \cong \mathbb{R}^n$). Pitfall: loops can be space-filling, so "homotope to miss a point" needs an argument (Hatcher uses Lebesgue numbers to make the loop piecewise linear near a point, or use van Kampen with two open hemispheres).

2.3 Seifert–van Kampen

The **free product** $G*H$ is the group of reduced words in $G \sqcup H$; it is the coproduct in the category of groups. An amalgamated product $G *_K H$ is $G * H$ modulo relations $i_1(k) = i_2(k)$.

Theorem (van Kampen). Let $X = \bigcup_\alpha A_\alpha$ with each A_α open, path-connected, containing x_0. If each $A_\alpha \cap A_\beta$ is path-connected, then $\Phi : *_\alpha \pi_1(A_\alpha) \to \pi_1(X)$ is surjective. If also each triple intersection is path-connected, $\ker \Phi$ is the normal subgroup N generated by $i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap A_\beta)$. So

$$\pi_1(X) \cong *_\alpha \pi_1(A_\alpha) \big/ N.$$

For two sets: $\pi_1(U \cup V) \cong \pi_1(U) *_{\pi_1(U \cap V)} \pi_1(V)$.

Proof idea. Surjectivity: subdivide a loop so each piece lies in one A_α and connect the subdivision points to x_0 by paths inside intersections. Injectivity modulo N: subdivide a nullhomotopy $I \times I$ into small rectangles, each mapping into one A_α, and move across the grid one rectangle at a time; each move is either a relation inside one $\pi_1(A_\alpha)$ or a relation in N. The triple-intersection condition handles grid vertices touching three or more sets.

Consequences.

- $\pi_1(\bigvee_\alpha S^1)$ is free on the circles (need each wedge point to have a contractible neighborhood: true for CW complexes).
- **Attaching 2-cells.** If Y is obtained from path-connected X by attaching 2-cells along loops φ_α, then $\pi_1(Y) \cong \pi_1(X)/N$, where N is the normal closure of the $[\varphi_\alpha]$ (after connecting to the basepoint). Attaching n-cells for $n \geq 3$ does not change π_1.
- Hence for any 2-dimensional CW complex with one 0-cell, π_1 has a presentation with one generator per 1-cell and one relator per 2-cell. **Every group is the fundamental group of some 2-dimensional CW complex.**
- Surfaces: $\pi_1(M_g) = \langle a_1, b_1, \dots, a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] \rangle$; abelianization $\mathbb{Z}^{2g}$, so the M_g are pairwise non-homotopy-equivalent.
- $\pi_1(\mathbb{RP}^2) = \mathbb{Z}/2$; Klein bottle K: $\langle a, b \mid abab^{-1} \rangle$.

Pitfall. The sets in van Kampen must be *open* (or deformation retract from open neighborhoods). Using closed sets like two closed hemispheres naively can fail for wild spaces.

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3. Covering spaces

3.1 Definitions and lifting

A **covering space** of X is a map $p : \tilde{X} \to X$ such that each $x \in X$ has an open neighborhood U with $p^{-1}(U)$ a disjoint union of open sets each mapped homeomorphically onto U ("evenly covered"). The sheets can be empty, so we usually require $\tilde{X}$ path-connected later.

Homotopy lifting property. Given a homotopy $f_t : Y \to X$ and a lift $\tilde{f}_0$ of f_0, there is a unique homotopy $\tilde{f}_t$ lifting f_t that starts at $\tilde{f}_0$.

Corollaries.

- $p_* : \pi_1(\tilde{X}, \tilde{x}_0) \to \pi_1(X, x_0)$ is **injective**; its image consists of loops whose lifts at $\tilde{x}_0$ are loops.
- For path-connected $\tilde{X}$, X, the number of sheets equals the index $[\pi_1(X, x_0) : p_* \pi_1(\tilde{X}, \tilde{x}_0)]$.

Lifting criterion. Let Y be path-connected and locally path-connected, $f : (Y, y_0) \to (X, x_0)$. A lift $\tilde{f} : (Y, y_0) \to (\tilde{X}, \tilde{x}_0)$ exists iff $f_* \pi_1(Y, y_0) \subseteq p_* \pi_1(\tilde{X}, \tilde{x}_0)$.

Proof sketch. Define $\tilde{f}(y)$ as the endpoint of the lift of $f \circ \gamma$ for a path γ from y_0 to y. The subgroup condition makes this independent of γ; local path-connectedness gives continuity.

Counterexample showing local path-connectedness is needed. The "quasi-circle" Y (a closed topologist's sine curve with its ends joined by an arc) is simply connected, but the natural map $Y \to S^1$ does not lift to $\mathbb{R}$.

Unique lifting. Two lifts $Y \to \tilde{X}$ of the same map that agree at one point of a connected Y agree everywhere.

3.2 Classification

X is **semi-locally simply connected** if each point has a neighborhood U such that $\pi_1(U, x) \to \pi_1(X, x)$ is trivial. CW complexes and manifolds qualify; the Hawaiian earring does not.

Theorem (universal cover). If X is path-connected, locally path-connected and semi-locally simply connected, then X has a simply connected covering space $\tilde{X}$. Construction: $\tilde{X} = \{\text{homotopy classes rel endpoints of paths starting at } x_0\}$, $p[\gamma] = \gamma(1)$, with a suitable topology.

Theorem (classification; Galois correspondence). For such X:

- Basepoint-preserving isomorphism classes of path-connected covers $(\tilde{X}, \tilde{x}_0) \to (X, x_0)$ correspond bijectively to subgroups $H \leq \pi_1(X, x_0)$, via $\tilde{X} \mapsto p_*\pi_1(\tilde{X}, \tilde{x}_0)$.
- Ignoring basepoints, isomorphism classes of covers correspond to **conjugacy classes** of subgroups.

Build the cover for H as $\tilde{X}_H = \tilde{X}/H$ (quotient of the universal cover by H acting by deck transformations).

3.3 Deck transformations

A **deck transformation** is a homeomorphism $\tau : \tilde{X} \to \tilde{X}$ with $p\tau = p$. They form a group $G(\tilde{X})$. A cover is **normal** (regular, Galois) if $G(\tilde{X})$ acts transitively on each fibre; equivalently $H = p_*\pi_1(\tilde{X})$ is normal in $\pi_1(X)$.

Theorem. $G(\tilde{X}) \cong N(H)/H$, where $N(H)$ is the normalizer. In particular for a normal cover $G(\tilde{X}) \cong \pi_1(X)/H$, and for the universal cover $G(\tilde{X}) \cong \pi_1(X)$.

Covering space actions. An action of G on Y is a *covering space action* if each y has a neighborhood U with $gU \cap U = \emptyset$ for $g \neq e$. Then $Y \to Y/G$ is a normal cover with deck group G. If Y is simply connected (and path-connected, locally path-connected), then $\pi_1(Y/G) \cong G$.

Examples.

- $\mathbb{Z}^n$ acting on $\mathbb{R}^n$ by translation: $\pi_1(T^n) = \mathbb{Z}^n$.
- $\mathbb{Z}/2$ acting on S^n by ± 1: $\pi_1(\mathbb{RP}^n) = \mathbb{Z}/2$ for $n \geq 2$.
- $\mathbb{Z}/m$ acting on $S^3 \subset \mathbb{C}^2$ by $(z, w) \mapsto (\zeta z, \zeta^q w)$, $\zeta = e^{2\pi i/m}$, $\gcd(q, m) = 1$: lens space $L(m, q)$ with $\pi_1 \cong \mathbb{Z}/m$.
- Covers of $S^1 \vee S^1$ are exactly graphs where each vertex has one incoming and one outgoing a-edge and b-edge. This gives the topological proof that **subgroups of free groups are free** and the **Schreier index formula**: a subgroup of index k in F_n is free of rank $k(n-1)+1$.

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4. Homology

4.1 Chain complexes

A **chain complex** $(C_\bullet, \partial)$ is a sequence of abelian groups with $\partial_n : C_n \to C_{n-1}$ and $\partial\partial = 0$. Its homology is $H_n = \ker \partial_n / \operatorname{im} \partial_{n+1}$. A **chain map** $f_\#$ commutes with ∂ and induces f_* on homology. A **chain homotopy** is $P : C_n \to D_{n+1}$ with $\partial P + P\partial = f_\# - g_\#$; chain homotopic maps induce the same map on homology.

Zig-zag lemma. A short exact sequence of chain complexes $0 \to A_\bullet \to B_\bullet \to C_\bullet \to 0$ gives a long exact sequence

$$\cdots \to H_n(A) \to H_n(B) \to H_n(C) \xrightarrow{\partial} H_{n-1}(A) \to \cdots$$

The connecting map: lift a cycle c to $b \in B_n$, then ∂b comes from a unique cycle $a \in A_{n-1}$; set $\partial[c] = [a]$. Do this diagram chase yourself once.

Five lemma. In a commutative ladder of exact rows, if the four outer maps are isomorphisms, so is the middle one.

4.2 Simplicial and singular homology

For a Δ-complex X, $\Delta_n(X)$ is free abelian on the n-simplices and

$$\partial\sigma = \sum_{i=0}^{n} (-1)^i \sigma|_{[v_0,\ldots,\hat{v}_i,\ldots,v_n]}.$$

Singular homology uses all maps $\sigma : \Delta^n \to X$ as generators of $C_n(X)$ with the same formula. It is huge but obviously functorial and topologically invariant. **Reduced homology** $\tilde{H}_n$ uses the augmentation $C_0 \to \mathbb{Z}$; $\tilde{H}_0(X) \oplus \mathbb{Z} \cong H_0(X)$, and $\tilde{H}_n = H_n$ for $n > 0$.

Basic facts.

- $H_0(X)$ is free abelian on the path components.
- $H_n(\text{point}) = 0$ for $n > 0$.
- **Homotopy invariance:** $f \simeq g \Rightarrow f_* = g_*$. (Proof: the prism operator subdivides $\Delta^n \times I$ into $(n+1)$-simplices and gives a chain homotopy.)
- **Relative homology** $H_n(X, A)$ from $C_\bullet(X)/C_\bullet(A)$, with long exact sequence

$$\cdots \to H_n(A) \to H_n(X) \to H_n(X, A) \to H_{n-1}(A) \to \cdots$$

- **Excision.** If $\overline{Z} \subseteq \operatorname{int} A$, then $H_n(X \setminus Z, A \setminus Z) \to H_n(X, A)$ is an isomorphism. Equivalently, for A, B with interiors covering X, $H_n(B, A \cap B) \cong H_n(X, A)$. Proof uses barycentric subdivision to replace chains by "small" chains; this is the hardest technical proof of the course. Learn the statement and what it buys you.
- **Good pairs.** If A is a nonempty closed subspace that is a deformation retract of a neighborhood (true for CW pairs), then $H_n(X, A) \cong \tilde{H}_n(X/A)$.
- Simplicial and singular homology agree for Δ-complexes (five lemma plus induction on skeleta).

Hurewicz in degree 1. For path-connected X, the map $\pi_1(X) \to H_1(X)$ sending a loop to its singular 1-cycle induces $\pi_1(X)^{ab} \cong H_1(X)$.

4.3 Spheres, degree, and classical applications

Theorem. $\tilde{H}_k(S^n) = \mathbb{Z}$ for $k = n$ and 0 otherwise.

Proof. Use the good pair (D^n, S^{n-1}) with $D^n/S^{n-1} \cong S^n$. Since $\tilde{H}_*(D^n) = 0$, the long exact sequence gives $\tilde{H}_k(S^n) \cong H_k(D^n, S^{n-1}) \cong \tilde{H}_{k-1}(S^{n-1})$. Induct from S^0. $\square$

Corollaries.

- $\partial D^n = S^{n-1}$ is not a retract of D^n; **Brouwer's fixed point theorem** holds in every dimension.
- **Invariance of dimension.** $\mathbb{R}^m \cong \mathbb{R}^n$ implies $m = n$: local homology $H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \cong \tilde{H}_{k-1}(S^{n-1})$ is a topological invariant.

Degree. A map $f : S^n \to S^n$ induces multiplication by an integer $\deg f$ on $H_n(S^n) \cong \mathbb{Z}$. Properties:

1. $\deg \mathrm{id} = 1$; $\deg(fg) = \deg f \deg g$; homotopic maps have equal degree; a non-surjective map has degree 0.
2. A reflection in a hyperplane has degree -1.
3. The antipodal map $-\mathrm{id}$ on S^n is a composite of $n+1$ reflections, so has degree $(-1)^{n+1}$.
4. If f has no fixed points, $f \simeq -\mathrm{id}$ via $\dfrac{(1-t)f(x) - tx}{|(1-t)f(x) - tx|}$, so $\deg f = (-1)^{n+1}$.
5. If $f(x) \neq -x$ for all x, then $f \simeq \mathrm{id}$ and $\deg f = 1$.

Hairy ball theorem. S^n has a nowhere-zero continuous tangent vector field iff n is odd. If v is such a field (normalized), $f_t(x) = \cos(\pi t)x + \sin(\pi t)v(x)$ is a homotopy from id to $-\mathrm{id}$, forcing $(-1)^{n+1} = 1$. For $n = 2k-1$ odd, $v(x_1, \dots, x_{2k}) = (-x_2, x_1, \dots, -x_{2k}, x_{2k-1})$ works.

Local degree. If $f^{-1}(y) = \{x_1, \dots, x_m\}$ is finite, $\deg f = \sum_i \deg f|_{x_i}$, where the local degree is computed in $H_n(U_i, U_i \setminus x_i)$. For smooth maps this is the signed count of preimages of a regular value, matching the smooth degree in 3.5.

4.4 Mayer–Vietoris

If $X = \operatorname{int} A \cup \operatorname{int} B$, the short exact sequence $0 \to C_\bullet(A \cap B) \to C_\bullet(A) \oplus C_\bullet(B) \to C_\bullet(A+B) \to 0$ (where $C(A+B)$ are sums of chains in A or B, which compute $H_*(X)$ by the small-chains theorem) gives

$$\cdots \to H_n(A \cap B) \xrightarrow{(i_*, -j_*)} H_n(A) \oplus H_n(B) \xrightarrow{k_* + l_*} H_n(X) \xrightarrow{\partial} H_{n-1}(A \cap B) \to \cdots$$

Same with reduced homology when $A \cap B \neq \emptyset$. The boundary map takes a cycle $z = a + b$ to $[\partial a] = -[\partial b]$.

Example: Klein bottle. Write K as two Möbius bands A, B glued along their boundary circle; $A \cap B \simeq S^1$, and $A, B \simeq S^1$. The boundary circle of a Möbius band wraps twice around its core. In degree 1: $H_1(A \cap B) = \mathbb{Z} \to \mathbb{Z} \oplus \mathbb{Z}$, $1 \mapsto (2, -2)$, which is injective, so $H_2(K) = 0$. Then $H_1(K) \cong (\mathbb{Z} \oplus \mathbb{Z})/\langle (2,-2) \rangle \cong \mathbb{Z} \oplus \mathbb{Z}/2$ (basis $(1,-1), (0,1)$; the subgroup is $2 \cdot (1,-1)$), using that $\tilde{H}_0(A \cap B) = 0$.

4.5 Cellular homology and Euler characteristic

For a CW complex X:

- $H_k(X^n, X^{n-1})$ is free abelian on the n-cells for $k = n$ and 0 otherwise.
- $H_k(X^n) = 0$ for $k > n$, and $H_k(X^n) \cong H_k(X)$ for $k < n$.

The **cellular chain complex** is $C_n^{CW} = H_n(X^n, X^{n-1})$ with d_n the composite $H_n(X^n, X^{n-1}) \to H_{n-1}(X^{n-1}) \to H_{n-1}(X^{n-1}, X^{n-2})$. **Theorem:** $H_n^{CW}(X) \cong H_n(X)$.

Cellular boundary formula.

$$d_n(e^n_\alpha) = \sum_\beta d_{\alpha\beta}\, e^{n-1}_\beta,$$

where $d_{\alpha\beta}$ is the degree of $S^{n-1} \xrightarrow{\varphi_\alpha} X^{n-1} \to X^{n-1}/(X^{n-1} \setminus e^{n-1}_\beta) \cong S^{n-1}$. For $n = 1$, d_1 is the usual vertex difference. For $n = 2$, $d_{\alpha\beta}$ is the exponent sum of the letter β in the attaching word.

Example: $\mathbb{RP}^n$. One cell e^k for $0 \le k \le n$. The attaching map $S^{k-1} \to \mathbb{RP}^{k-1}$ followed by the collapse to S^{k-1} is a map whose local degrees at the two preimages are 1 and $\deg(-\mathrm{id}_{S^{k-1}}) = (-1)^k$. So $d_k = 1 + (-1)^k$: zero for k odd, 2 for k even. The cellular complex is

$$0 \to \mathbb{Z} \xrightarrow{d_n} \cdots \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0,$$

giving $H_0 = \mathbb{Z}$; $H_k = \mathbb{Z}/2$ for k odd, $0 < k < n$; $H_k = 0$ for k even, $0 < k \le n$; and $H_n = \mathbb{Z}$ if n odd.

Example: $\mathbb{CP}^n$. Cells only in even degrees, so all $d = 0$ and $H_{2k} = \mathbb{Z}$ for $0 \le k \le n$, zero otherwise.

Euler characteristic. For a finite CW complex, $\chi(X) = \sum_n (-1)^n c_n$ where c_n is the number of n-cells. **Theorem:** $\chi(X) = \sum_n (-1)^n \operatorname{rank} H_n(X)$. (Rank is additive along exact sequences of finitely generated abelian groups; apply to $0 \to Z_n \to C_n \to B_{n-1} \to 0$ and $0 \to B_n \to Z_n \to H_n \to 0$.) So χ is a homotopy invariant. $\chi(M_g) = 2 - 2g$, $\chi(N_h) = 2 - h$, $\chi(S^n) = 1 + (-1)^n$. For an n-sheeted cover $\tilde{X} \to X$ of a finite CW complex, $\chi(\tilde{X}) = n\,\chi(X)$ (lift the cells).

4.6 Homology with coefficients

Replace $C_n(X)$ by $C_n(X) \otimes G$. With $G = \mathbb{Z}/2$, orientation signs disappear: $H_k(\mathbb{RP}^n; \mathbb{Z}/2) = \mathbb{Z}/2$ for $0 \le k \le n$. Field coefficients make every homology group a vector space, and the Euler characteristic can be computed with any field.

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5. Cohomology and cup product (light)

5.1 Cohomology

Dualize: $C^n(X; G) = \mathrm{Hom}(C_n(X), G)$, $\delta\varphi = \varphi \circ \partial$, so $\delta : C^n \to C^{n+1}$. Then $H^n(X; G) = \ker \delta / \operatorname{im} \delta$. Cohomology is contravariant: $f : X \to Y$ gives $f^* : H^n(Y) \to H^n(X)$. It satisfies homotopy invariance, long exact sequences, excision and Mayer–Vietoris, all with arrows reversed.

Universal coefficient theorem. There is a split short exact sequence

$$0 \to \mathrm{Ext}(H_{n-1}(X), G) \to H^n(X; G) \to \mathrm{Hom}(H_n(X), G) \to 0.$$

For finitely generated groups, $\mathrm{Ext}(\mathbb{Z}, G) = 0$ and $\mathrm{Ext}(\mathbb{Z}/m, G) = G/mG$. Consequence with $G = \mathbb{Z}$: if $H_n = \mathbb{Z}^{b_n} \oplus T_n$ (torsion T_n), then $H^n(X; \mathbb{Z}) \cong \mathbb{Z}^{b_n} \oplus T_{n-1}$. Torsion moves up one degree. Example: $H^2(\mathbb{RP}^2; \mathbb{Z}) \cong \mathbb{Z}/2$ while $H_2(\mathbb{RP}^2) = 0$.

5.2 Cup product

For a commutative ring R, $\varphi \in C^k(X; R)$ and $\psi \in C^\ell(X; R)$,

$$(\varphi \smile \psi)(\sigma) = \varphi(\sigma|_{[v_0, \dots, v_k]})\, \psi(\sigma|_{[v_k, \dots, v_{k+\ell}]}).$$

Then $\delta(\varphi \smile \psi) = \delta\varphi \smile \psi + (-1)^k \varphi \smile \delta\psi$, so $\smile$ descends to $H^k \times H^\ell \to H^{k+\ell}$. This makes $H^*(X; R)$ a graded ring that is **graded commutative**: $a \smile b = (-1)^{k\ell} b \smile a$. Induced maps f^* are ring homomorphisms, so the ring structure is a homotopy invariant that is finer than the groups.

Rings to know (statements; Hatcher §3.2).

- $H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1})$, $|x| = 2$.
- $H^*(\mathbb{RP}^n; \mathbb{Z}/2) \cong \mathbb{Z}/2[x]/(x^{n+1})$, $|x| = 1$.
- $H^*(T^n; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}[x_1, \dots, x_n]$, $|x_i| = 1$ (exterior algebra).
- $H^*(M_g; \mathbb{Z})$: $H^1 = \mathbb{Z}^{2g}$ with basis α_i, β_i, $\alpha_i \smile \beta_i = -\beta_i \smile \alpha_i$ a generator of H^2, all other products of basis elements zero.
- Wedge sums: $\tilde{H}^*(X \vee Y) \cong \tilde{H}^*(X) \times \tilde{H}^*(Y)$ as rings, so products of classes from different summands vanish.

Example. $\mathbb{CP}^2$ and $S^2 \vee S^4$ have the same homology ($\mathbb{Z}$ in degrees $0, 2, 4$) but in $\mathbb{CP}^2$ the square of the degree-2 generator generates H^4, while in $S^2 \vee S^4$ it is zero. So they are not homotopy equivalent. The same argument shows the Hopf map $S^3 \to S^2$ is not nullhomotopic, because $\mathbb{CP}^2$ is S^2 with a 4-cell attached by the Hopf map.

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6. Common pitfalls

1. **Basepoints.** π_1 is only defined up to non-canonical isomorphism when you move the basepoint (canonical only when π_1 is abelian). Homotopies of maps can move the basepoint.
2. **Van Kampen hypotheses.** Open sets, path-connected intersections. Two overlapping arcs covering S^1 have disconnected intersection, so you cannot conclude $\pi_1(S^1) = 0$.
3. **Covering classification needs local niceness.** Without semi-local simple connectivity there may be no universal cover (Hawaiian earring).
4. **Signs in boundary maps.** Always check $\partial\partial = 0$ on an example. In cellular homology, compute degrees of attaching maps carefully, especially with orientation-reversing identifications ($\mathbb{RP}^n$, Klein bottle).
5. **Homology does not detect everything.** The Poincaré homology sphere has the homology of S^3 but nontrivial π_1. $\mathbb{CP}^2$ and $S^2 \vee S^4$ have equal homology but are distinguished by cup products.
6. **Contravariance.** Cohomology reverses arrows; the connecting map in Mayer–Vietoris goes $H^n(A \cap B) \to H^{n+1}(X)$.
7. **Torsion and Hom.** $\mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}) = 0$: torsion in H_n is invisible in $\mathrm{Hom}(H_n, \mathbb{Z})$ and reappears in H^{n+1} via Ext.
8. H_1 **is the abelianization, not** π_1**.** $S^1 \vee S^1$ and T^2 minus a point both have $H_1 = \mathbb{Z}^2$; they are in fact homotopy equivalent, but T^2 itself is not homotopy equivalent to $S^1 \vee S^1$ (H_2 differs).

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7. Connections forward

- **Smooth manifolds (3.5).** de Rham's theorem: $H^k_{dR}(M) \cong H^k(M; \mathbb{R})$. Smooth degree equals homological degree. Orientation can be defined via local homology $H_n(M, M \setminus x)$, and the orientation double cover is a covering space.
- **Poincaré duality** (Hatcher §3.3): for a closed orientable n-manifold, $H^k(M) \cong H_{n-k}(M)$. The natural next step after this module.
- **Higher homotopy groups and fibrations** (Hatcher Ch. 4): long exact sequence of a fibration, Hurewicz theorem, Whitehead's theorem.

- **Complex analysis (2.2).** Winding numbers are degrees of maps to S^1; Riemann surfaces of multivalued functions are covering spaces; monodromy is the action of π_1 on fibres.
- **Algebra.** Group cohomology is cohomology of $K(G,1)$; covering theory mirrors Galois theory; the étale fundamental group in algebraic geometry is a direct analogue (algebraic geometry elective).
- **Lie groups (elective).** $\pi_1(SO(3)) = \mathbb{Z}/2$ with universal cover $SU(2) \cong S^3$.